Credit Default Swaptions

Seminar Module

conducted by

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What is a Credit Default Swaption?

Definition: A Credit Default Swaption is:

- An option on a credit derivatives index (e.g. CDX) or on a single-name CDS
- Analogous to an interest rate swaption

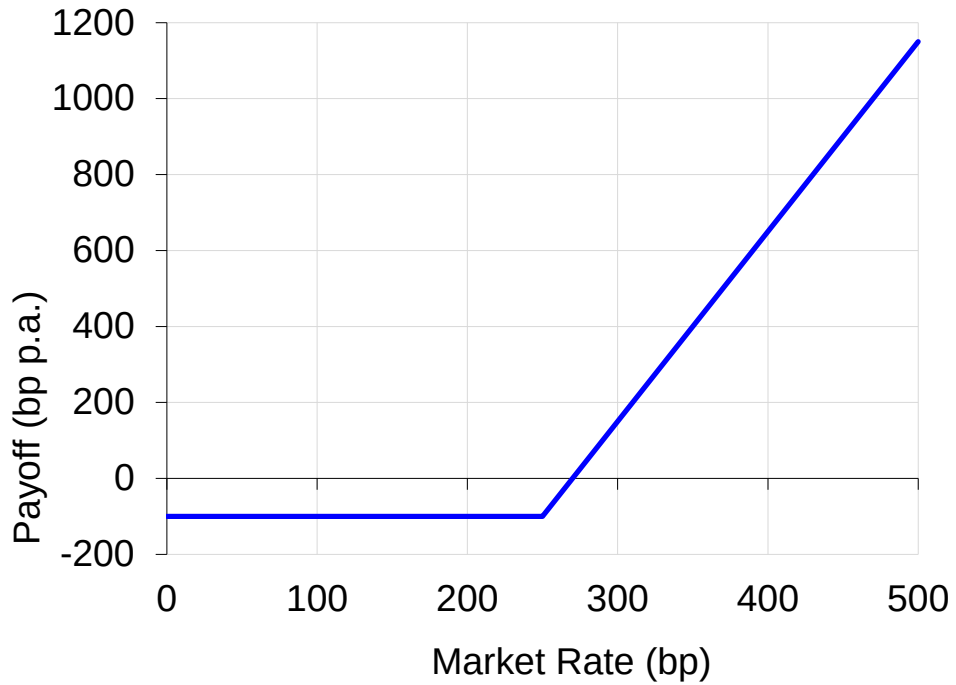
Market Conventions

- The Credit Default Swaption buyer has the right to transact the underlying CDX:
  - At a given premium rate (the strike)
  - For a given maturity
  - On or before a specific date in the future (the maturity date)
- Call / Payers swaption – the right to buy protection
- Put / Receivers swaption – the right to sell protection
- Credit Default Swaption premiums:
  - Are normally quoted as a percentage of the notional principal
  - Are normally paid up-front
- European-style physical exercise is the most common

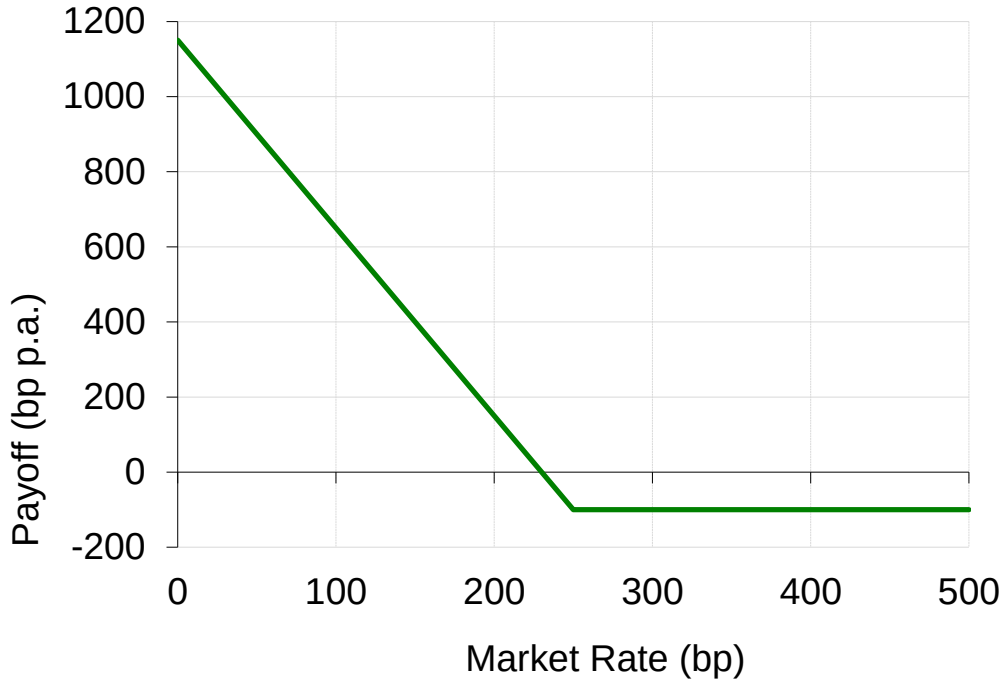
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Payoff Profiles

Call swaption



Put swaption



Assumptions:

- No credit events occur
- Swaption exercises into a 5yr CDX

- Premium: 100 bp up-front
- Strike: 250 bp p.a.

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What Happens if a Credit Event Occurs?

Index Swaptions

- The buyer of a payer option (or seller of a receiver) can trigger a credit event on option exercise
- If exercised, the buyer of a payer swaption receives any default losses
  - It may therefore be profitable for the buyer to exercise a payer even if the index spread is below the strike
  - Defaults thus **increase** the range over which a payer can be profitably exercised
- If exercised, the seller of a receiver swaption also receives any default losses
- It may therefore be **less** profitable for the buyer to exercise a receiver because the buyer has to pay default losses
- Defaults thus **decrease** the range for profitable exercise of a receiver

Single-Name Credit Default Swaptions

- Credit Default Swaptions knock out
- Credit Default Swaptions provide protection against spread risk, not default risk

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Credit Default Swaption Applications

- Example 1: Creating a Cancellable CDX
  - Buying a 5yr CDX, and
  - Buying a 2yr put Credit Default Swaption on a 3yr CDX
  - … creates a five-year CDX cancellable after two years
- Example 2: Creating an Extendable CDX
  - Buying a 5yr CDX, and
  - Buying a 5yr call Credit Default Swaption on a 2yr CDX
  - … creates a five-year CDX extendable to seven years
- Example 3: Creating a synthetic CLN
  - Buy a high-grade 4yr FRN yielding LIBOR+5p
  - Sell a put swaption on entity ABC plc for premium of 40bp
  - … creates a CLN with an enhanced yield of LIBOR+45bp

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Cancellable CDX

- A cancellable CDX is a combination of:
  - A regular CDX plus
  - A Credit Default Swaption
- A cancellable long CDX is:
  - Long a regular CDX (buying protection)
  - Long a put Credit Default Swaption (giving the right to sell protection, cancelling the long CDX)
- A cancellable short CDX is:
  - Short a regular CDX (selling protection)
  - Long a call Credit Default Swaption (giving the right to buy protection, cancelling the short CDX)

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Credit Default Swaption Strategies

- Buying OTM payer swaptions to protect a long credit portfolio in the event of dramatic credit spread widening
  - Many portfolio managers prefer this strategy compared to buying the underlying index protection because their maximum downside is limited to the upfront option premium paid
- Writing a payer swaption to reduce the cost of holding a short position in the underlying index
  - Investors using this strategy reduce the cost of the index short but give up some of the upside from spread widening
  - Note, however, that if the swaption is exercised, the option writer loses the short index position and does not benefit from any future spread widening
- Express views on CDX index volatility
  - If implied swaption volatility is **higher** than daily realized volatility of the underlying index, consider **selling** a straddle
  - If implied swaption volatility is **lower** than daily realized volatility of the underlying index, consider **buying** a straddle
  - Cross-market traders can express relative value views between credit and equity volatility

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Pricing European-Style Credit Default Swaptions

- One approach is to price these using a modified Black model
  (Assumes that the forward rate of the underlying CDX is lognormally distributed)

$$C = \left[s \cdot N(d_1) - X \cdot N(d_2) \right] \cdot \sum_{i=t}^{t+T} v_i \qquad P = \left[X \cdot N(-d_2) - s \cdot N(-d_1) \right] \cdot \sum_{i=t}^{t+T} v_i$$

$$d_1 = \frac{\ln\left(\frac{s}{X}\right) + \frac{\sigma^2 t}{2}}{\sigma\sqrt{t}} \qquad d_2 = d_1 - \sigma\sqrt{t}$$

where:

- C is the premium rate for a call Credit Default Swaption
- P is the premium rate for a put Credit Default Swaption
- s is the underlying (deferred-start) CDX rate
- σ is the volatility of the underlying (deferred-start) swap rate
- t is the length of time until the Credit Default Swaption expires (in years)
- T is the length of time covered by the underlying (deferred-start) CDX
- X is the strike rate of the swaption

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Pricing American and Bermudan Swaptions

- Need to consider the stochastic process of the entire CDX term-structure…
- … Not just the single forward CDX rate
- Alternatives:
  - One-factor term-structure models
  - Multi-factor term-structure models
  - References:
    - Jarrow, R., D. Lando, and S. Turnbull, 1997, “A Markov Model for the Term Structure of Credit Spreads,” Review of Financial Studies 10, 481-523.
    - Tucker, A., and J. Wei, 2005, “Credit Default Swaptions”, Journal of Fixed Income, Vol 15, No. 1
    - Jamshidian, F, 2002, “Valuation of Credit Default Swap and Swaptions”, unpublished paper

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